

# Git & GitHub Revision Master Guide

Simple, Practical & Complete — Concepts, Commands, Diagnostics, Viva & Teaching Script

## 1. The Big Picture in Plain English

Think of it like this: **Your PC** → **Git** → **GitHub**

| Part                    | What It Actually Does (Simple Words)                                                                                                                                                                                                                      |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Git</b>              | A software installed on your local computer that acts like a time-machine. It tracks changes, saves checkpoints (commits), and remembers project history.                                                                                                 |
| <b>GitHub</b>           | A website / cloud service where you host copies of your Git repositories so they are safe online and shareable.                                                                                                                                           |
| <b>origin</b>           | Just a nickname for your GitHub project link. It saves you from typing the full URL every time you send code.                                                                                                                                             |
| <b>Fetch vs Pull</b>    | <code>git fetch</code> downloads updates from GitHub to check them without touching your files. <code>git pull</code> downloads AND merges changes into your work immediately.                                                                            |
| <b>Identity vs Auth</b> | <code>user.name</code> / <code>user.email</code> is just a <i>signature/stamp</i> on your commits. Git Credential Manager (GCM) handles your actual <i>login &amp; password/token</i> into GitHub. Changing your email never logs you into a new account! |

## 2. Local vs Global Identity (Separate College & Personal Work)

By default, Git uses your **Global** identity everywhere. But if you have separate accounts (e.g., College projects vs Personal ML projects), configure the identity **Locally** inside each project folder:

| Config Level           | Command                                                                                                                     | Where It Applies                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Global</b>          | <code>git config --global user.name "Your Name"</code><br><code>git config --global user.email "name@gmail.com"</code>      | Every repo on your computer unless overridden.                               |
| <b>Local</b>           | <code>git config --local user.name "College Name"</code><br><code>git config --local user.email "rollno@college.edu"</code> | Only the single folder/repo you are currently in. Overrides global settings. |
| <b>Debug / Inspect</b> | <code>git config --show-origin --get user.email</code><br><code>git config --list --show-origin</code>                      | Shows exact value AND tells you which configuration file supplied it.        |

## 3. Step-by-Step Practical Setup From Scratch

1. Open Terminal in your project folder: `cd "A:\Machine Learning"`
2. Check if Git is already active: `git status` (If it says `fatal: not a git repository`, run `git init`).
3. Set your identity for this project:

```
git config --local user.name "Aman Raj"
git config --local user.email "aman@example.com"
```

4. Connect to GitHub (Remote):

```
# First time linking:
git remote add origin "https://github.com/USERNAME/REPO.git"

# If already linked or URL is wrong:
git remote set-url origin "https://github.com/USERNAME/REPO.git"
```

5. Verify connection before pushing (Safe Probe):

```
git fetch origin
```

**Why fetch first?** It tests whether the URL is right and whether your GitHub login works, without modifying or breaking any of your local files.

## 4. Diagnostics, Errors & Windows Credential Manager

| Error / Problem                                   | Why It Happens                                                                                                                                 | Exact Fix                                                                                                                                                     |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>fatal: not a git repository</code>          | You are in the wrong folder, or <code>git init</code> was never run.                                                                           | Navigate to correct folder or run <code>git init</code> .                                                                                                     |
| <code>fatal: remote origin already exists</code>  | You tried running <code>remote add</code> again on an already linked project.                                                                  | Use <code>git remote set-url origin "&lt;URL&gt;"</code> .                                                                                                    |
| <code>remote: Repository not found</code>         | Mistyped URL OR you are trying to access a <b>private repo</b> without proper account access. (GitHub pretends it doesn't exist for security). | 1. Check URL with <code>git remote -v</code><br>2. Confirm the GitHub user logged into Git has collaborator access.                                           |
| <code>gh : The term 'gh' is not recognized</code> | GitHub CLI tool isn't installed.                                                                                                               | <b>Safe to ignore!</b> You do not need GitHub CLI ( <code>gh</code> ). Pure Git commands work completely fine.                                                |
| Wrong account being used to push                  | Windows saved an old login session.                                                                                                            | Open <b>Windows Search</b> → <b>Credential Manager</b> → <b>Windows Credentials</b> . Find <code>git:https://github.com</code> and remove it to log in fresh. |

## 5. The 4 Golden Questions & Core Cheat Sheet

Whenever Git confuses you, run these 4 commands in order:

| Step | Question to Ask Yourself         | Command to Run                                                         |
|------|----------------------------------|------------------------------------------------------------------------|
| 1    | Where am I & what is changed?    | <code>git status</code>                                                |
| 2    | Where is this project connected? | <code>git remote -v</code>                                             |
| 3    | Who am I signing my commits as?  | <code>git config user.name</code> & <code>git config user.email</code> |
| 4    | Can Git actually talk to GitHub? | <code>git fetch origin</code>                                          |

## 6. 5-Minute Teaching Script & Viva / Exam Prep

**How to explain this to a classmate in 30 seconds:**

"Git is the save button on your local PC; GitHub is the cloud storage. `origin` is just the shortcut name for the link. If you want separate college and personal identities, configure them with `git config --local`. And if private repo access fails, test it safely with `git fetch origin` before pushing."

### Viva / Interview Rapid Fire

**Q1. Does changing git user.email authenticate you to a new GitHub account?**

**No.** `user.email` only writes your name/email into the commit history. Git Credential Manager (GCM) handles login authorization.

**Q2. Why does a private repository say "Repository not found" instead of "Access Denied"?**

For security and privacy, GitHub will not disclose whether a private repository exists to an unauthorized user.

**Q3. What is the difference between git fetch and git pull?**

`git fetch` downloads commits and branch info from the remote without touching your working files. `git pull` fetches AND merges changes immediately.

**Q4. What is the role of `--show-origin`?**

It traces the exact physical configuration file path (system, global, or local) from which Git read that specific setting.